

Theoretical Computer Science Exam, September 14th, 2017

The exam consists of **4 exercises**. Available time: 1 hr 45 min.; you may write your answers in Italian, if you wish.

All exercises must be addressed for the written exam to be considered sufficient.

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Exercise 1 (9 pts)

Design an automaton that accepts, and a grammar that generates, the languages defined below; both the automata and the grammar must be of the least powerful type.

- 1) $L1 = \{ w \in \{a, b\}^* \mid (\#_a(w) + 2 \cdot \#_b(w)) \bmod 5 = 0 \}$
where $\#_a(w)$ and $\#_b(w)$ denote the number of symbols a and b occurring in string w , and MOD denotes the integer division remainder operator. For instance, $abb \in L1$, $\varepsilon \in L1$, $aba \notin L1$.

In addition, write the computation of the automaton, and a derivation of the grammar, that respectively accept and generate the string abb .

- 2) $L2 = \{ a^n b^m c^k \mid m, n, k \geq 0 \text{ and } m = n + k \}$
For instance, $a^2 b^5 c^3 \in L2$, $\varepsilon \in L2$, $a^2 b^4 c^3 \notin L2$.

Exercise 2 (8 pts)

Preliminary part

1. Two natural numbers a and b are said to be *mutually prime* if their greatest common divisor is 1. Formalize this notion by means of a binary predicate

$$\text{mutuallyPrime}(a, b)$$

to be specified by means of a formula in mathematical logics that uses the customary predicates and operators of integer arithmetic on natural numbers.

2. Consider a subprogram

computePrimes (v, n, xnp, x, y)

where v and n are input parameters, while xnp, x and y are output parameters; v is an array of integer numbers having $n > 1$ elements that are all distinct; parameter xnp is boolean and after execution it is true if and only if there exists a pair of elements in the array that are not mutually prime; if this is the case, the output parameters x and y are set equal to the positions of a pair of array elements ($v[x], v[y]$) such that $\text{mutuallyPrime}(v[x], v[y])$ does not hold (otherwise x and y are undefined). For instance, if $v = [3, 5, 7, 11, 8, 13]$ (hence $n = 6$) then $xnp = \text{false}$ and the value of x and y is undefined. As another instance, if $v = [3, 5, 10, 9, 12, 8]$ (hence $n = 6$) then $xnp = \text{true}$, and one may have $x = 2$ and $y = 3$ (we assume array indices starting from 1), corresponding to the pair of elements (5, 10) not being mutually prime.

Specify subprogram *computePrimes* by means of pre- and post-conditions; to this end you may use the previously specified *mutuallyPrime* predicate.

3. (Optional) Modify the specification of the previous point so that parameter xnp is true if and only if there exists a pair of elements in the array that are not mutually prime, but in this case the output parameters x and y are set equal to the positions of a pair of array elements such that they are not mutually prime, and there does not exist any other different pair of array elements having the same property of not being mutually prime, and such that their product is greater than that of $v[x]$ and $v[y]$. For instance, if $v = [3, 5, 10, 9, 12, 8]$ then $xnp = \text{true}$, $x = 3$ and $y = 5$ corresponding to the pair of elements (10, 12), because all other pairs of array elements that are not mutually prime are such that their product is less than $120 = 10 \cdot 12$ (for instance, another pair of elements that are not mutually prime is (12, 9) but their product is less than 120).

Exercise 3 (8 pts)

Preliminarily, prove that the union of two semidecidable sets, S_1 and S_2 , is also semidecidable. Your argument should be as formal and precise as possible.

Prove the following property: the difference set, $D - S$, between any decidable set D and any semidecidable set S strictly included in D (i.e., such that $S \subset D$ and $S \neq D$) is *not* necessarily semidecidable.

Exercise 4 (8 pts)

Consider language $L = \{ a^n b^m c^k \mid m, n, k \geq 1 \text{ and } k = 2 \cdot n \text{ and } m = n + k \}$.

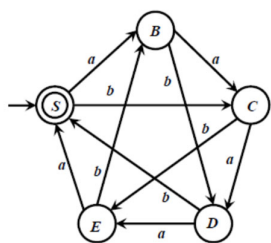
Design a Turing machine (preferably having only one memory tape) that recognizes the language with complexity class $\Theta(n)$. For this machine, determine in a precise way the time complexity $T(n)$ as a function of the input string length.

Sketch a single tape Turing machine that recognizes the language with a minimal complexity class, providing an adequate justification for your reply.

Solutions

Exercise 1

1.



Computation accepting abb : $S \xrightarrow{a} B \xrightarrow{b} D \xrightarrow{b} S$

$S \rightarrow \varepsilon$

$S \rightarrow a B \mid b C$

$B \rightarrow a C \mid b D$

$C \rightarrow a D \mid b E$

$D \rightarrow a E \mid b S$

$E \rightarrow a S \mid b B$

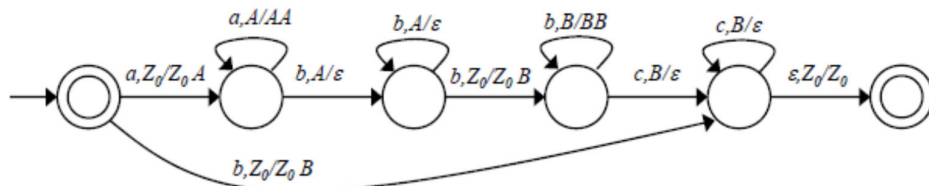
Derivation generating abb : $S \Rightarrow aB \Rightarrow abD \Rightarrow abbS \Rightarrow abb$

2.

$S \rightarrow A C$

$A \rightarrow a A b \mid \varepsilon$

$C \rightarrow b C c \mid \varepsilon$



Exercise 2

$$1. \text{ mutuallyPrime}(a, b) \leftrightarrow \neg \exists q_a, q_b, d (d > 1 \wedge a = d \cdot q_a \wedge b = d \cdot q_b)$$

2.

$$\text{pre: } n > 1 \wedge \neg \exists i, j (1 \leq i < j \leq n \wedge v[i] = v[j])$$

$$\text{post: } xnp \rightarrow 1 \leq x < y \leq n \wedge \neg \text{mutuallyPrime}(v[x], v[y])$$

3.

$$\text{post: } xnp \rightarrow 1 \leq x < y \leq n \wedge \neg \text{mutuallyPrime}(v[x], v[y]) \wedge$$

$$\neg \exists w, z (1 \leq w < z \leq n \wedge ((w \neq x \wedge w \neq y) \vee (z \neq x \wedge z \neq y)) \wedge$$

$$\neg \text{mutuallyPrime}(v[w], v[z]) \wedge v[w] \cdot v[z] > v[x] \cdot v[y])$$

Exercise 3

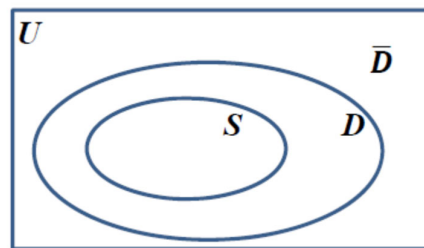
Preliminarily: yes, the union of the two semidecidable sets S_1 and S_2 is semidecidable.

In fact, if we call gs_1 and gs_2 the generating functions of S_1 and S_2 , then any total computable function having as *image* the union of the two generating functions gs_1 and gs_2 would be a generating function for $S_1 \cup S_2$. Therefore the generating function for set $S_1 \cup S_2$ exists.

Another argument consist of defining the enumeration of $S_1 \cup S_2$ based on that of S_1 and S_2 . Let $enum_1 = \{n1_0, n1_1, n1_2, \dots\}$ be an enumeration of S_1 , and $enum_2 = \{n2_0, n2_1, n2_2, \dots\}$ be an enumeration of S_2 ; then an enumeration $enum_{12}$ of $S_1 \cup S_2$ is defined as follows $enum_{12} = \{n1_0, n2_0, n1_1, n2_1, n1_2, n2_2, \dots\}$.

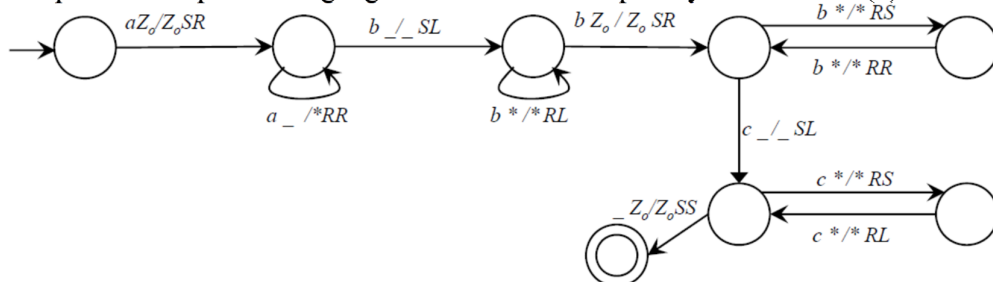
Proof of the stated property.

Since D is decidable, its complement $\bar{D}$ is also decidable, hence semidecidable. If $D - S$ was semidecidable, then also $(D - S) \cup \bar{D}$, would be semidecidable, being the union of two semidecidable sets. But the set $(D - S) \cup \bar{D}$ is equal to the complement of S . Hence, being both S and $\bar{S}$ semidecidable, S would be decidable, contradicting the hypothesis that it is semidecidable.



Exercise 4

The following 1-tape TM accepts the language with the time complexity function is $T(n) = n + 4$.



The same language can be accepted by a single tape Turing machine with complexity class $\Theta(n^2)$, because the machine necessarily must go up and down the string, canceling the symbols, a number of times that is proportional to the string length.